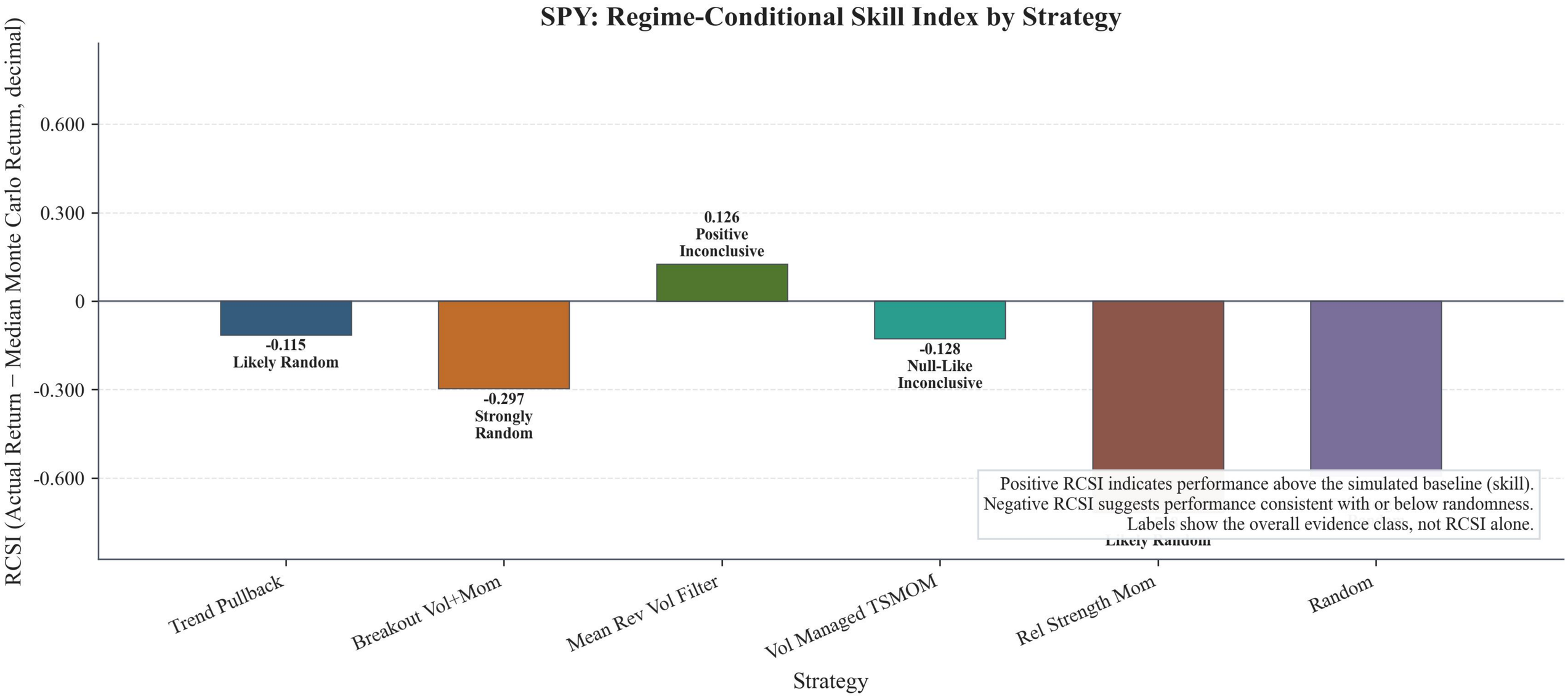
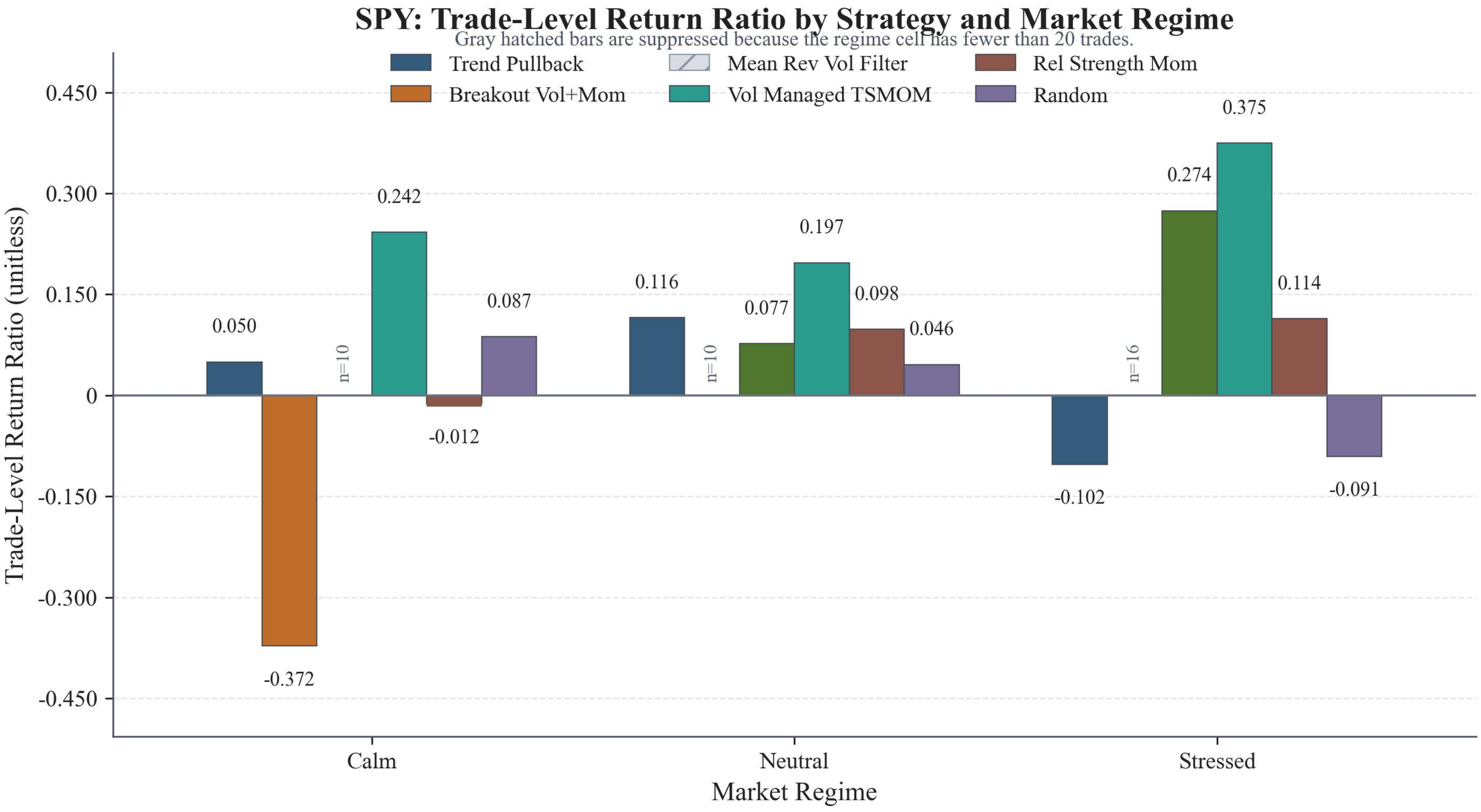
SPY: Strategy Evidence and Confidence Overview

Evidence labels use scale-free statistics: p-value, percentile, normalized RCSI, robustness, and stability.

Strategy	Evidence	Verdict	Confidence	p-value	Prom.	Pctile	RCSI_z
Trend Pullback	Likely Random	Luck	Moderate	0.665	0.18	33.5	-0.50
Breakout Vol+Mom	Strongly Random	Luck	High	0.886	0.05	11.4	-1.14
Mean Rev Vol Filter	Positive but Inconclusive	Inconclusive	Low	0.210	0.68	79.0	0.77
Vol Managed TSMOM	Null-Like / Inconclusive	Inconclusive	Low	0.552	0.26	44.8	-0.30
Rel Strength Mom	Likely Random	Luck	Moderate	0.839	0.08	16.1	-0.88
Random	Strongly Random	Luck	High	0.927	0.03	7.3	-1.21

Interpretation:
Prom. = -log10(p), so larger values indicate rarer outcomes under the null.
Pctile and RCSI_z provide scale-free evidence and are more comparable across tickers.
Benchmark context appears in the comparison table and equity curve rather than this verdict summary.


SPY: Regime-Conditional Skill Index by Strategy
RCSI (Actual Return − Median Monte Carlo Return, decimal)
0.600
0.300
0
-0.300
-0.600
-0.115
Likely Random
-0.297
Strongly
Random
0.126
Positive
Inconclusive
-0.128
Null-Like
Inconclusive
Likely Random
Positive RCSI indicates performance above the simulated baseline (skill).
Negative RCSI suggests performance consistent with or below randomness.
Labels show the overall evidence class, not RCSI alone.
Trend Pullback
Breakout Vol+Mom
Mean Rev Vol Filter
Vol Managed TSMOM
Rel Strength Mom
Random
Strategy


SPY: Trade-Level Return Ratio by Strategy and Market Regime
Gray hatched bars are suppressed because the regime cell has fewer than 20 trades.
Trend Pullback
Mean Rev Vol Filter
Rel Strength Mom
Breakout Vol+Mom
Vol Managed TSMOM
Random
Trade-Level Return Ratio (unitless)
0.450
0.300
0.150
0
-0.150
-0.300
-0.450
0.050
-0.372
n=10
0.242
-0.012
0.087
0.116
n=10
0.077
0.197
0.098
0.046
n=16
-0.102
0.274
0.375
0.114
-0.091
Calm
Neutral
Stressed
Market Regime

SPY: Strategy Performance Across Market Regimes

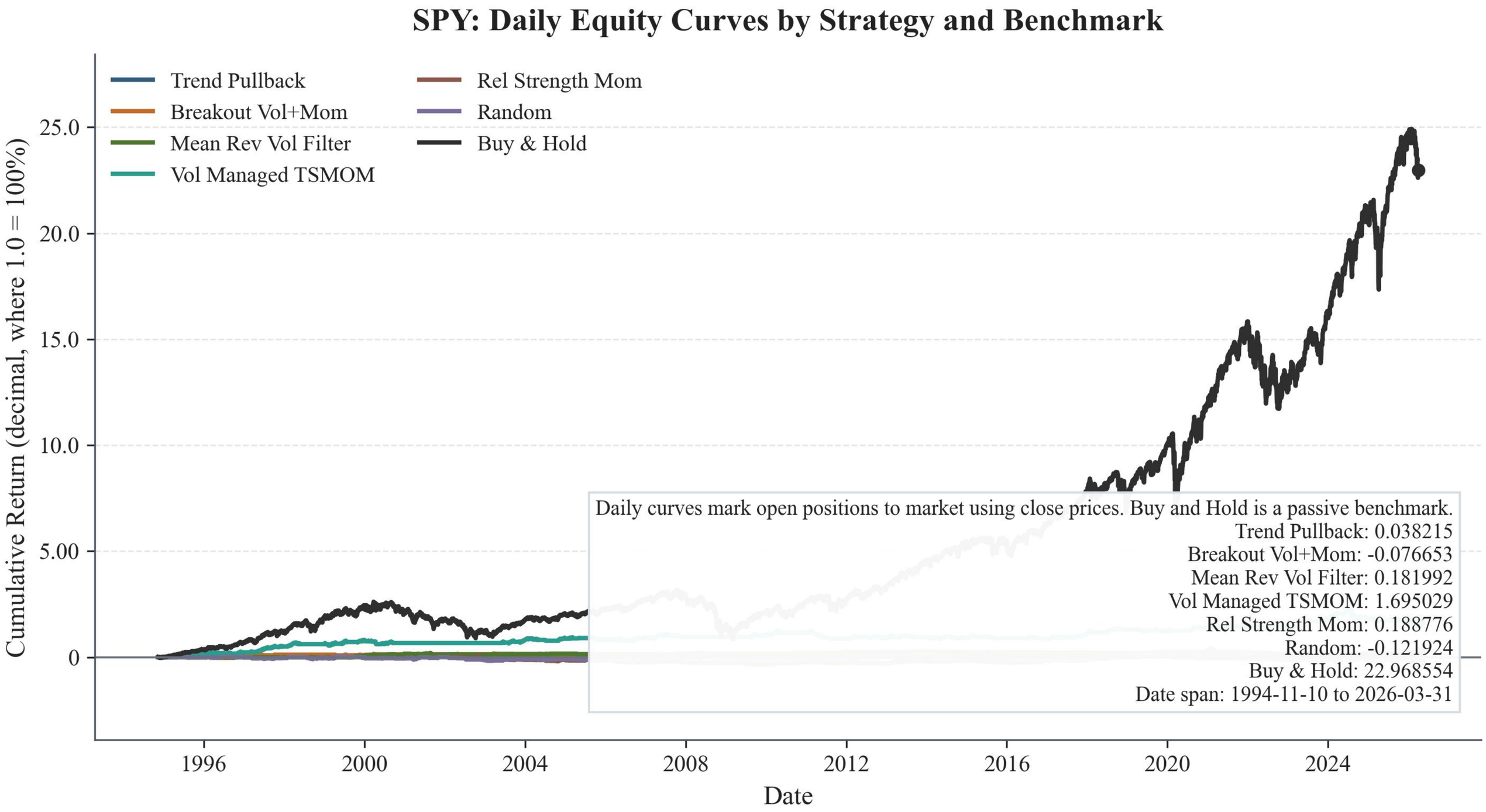
Gray cells are suppressed because the regime cell has fewer than 20 trades.

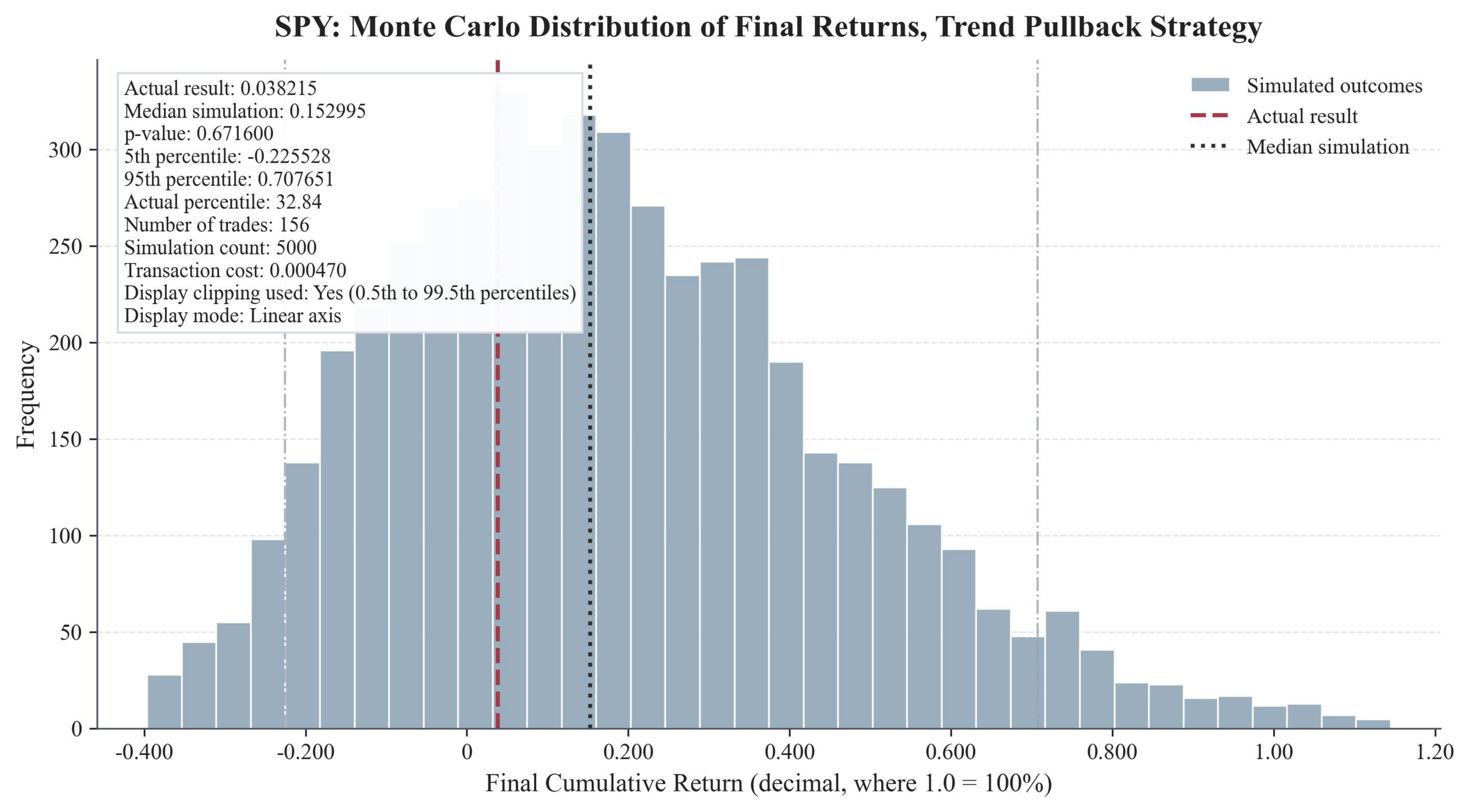
Strategy	Calm	Neutral	Stressed
Trend Pullback	0.05	0.12	-0.10
Breakout Vol+Mom	-0.37	Suppressed n=10	Suppressed n=16
Mean Rev Vol Filter	Suppressed n=10	0.08	0.27
Vol Managed TSMOM	0.24	0.20	0.37
Rel Strength Mom	-0.01	0.10	0.11
Random	0.09	0.05	-0.09

Market Regime

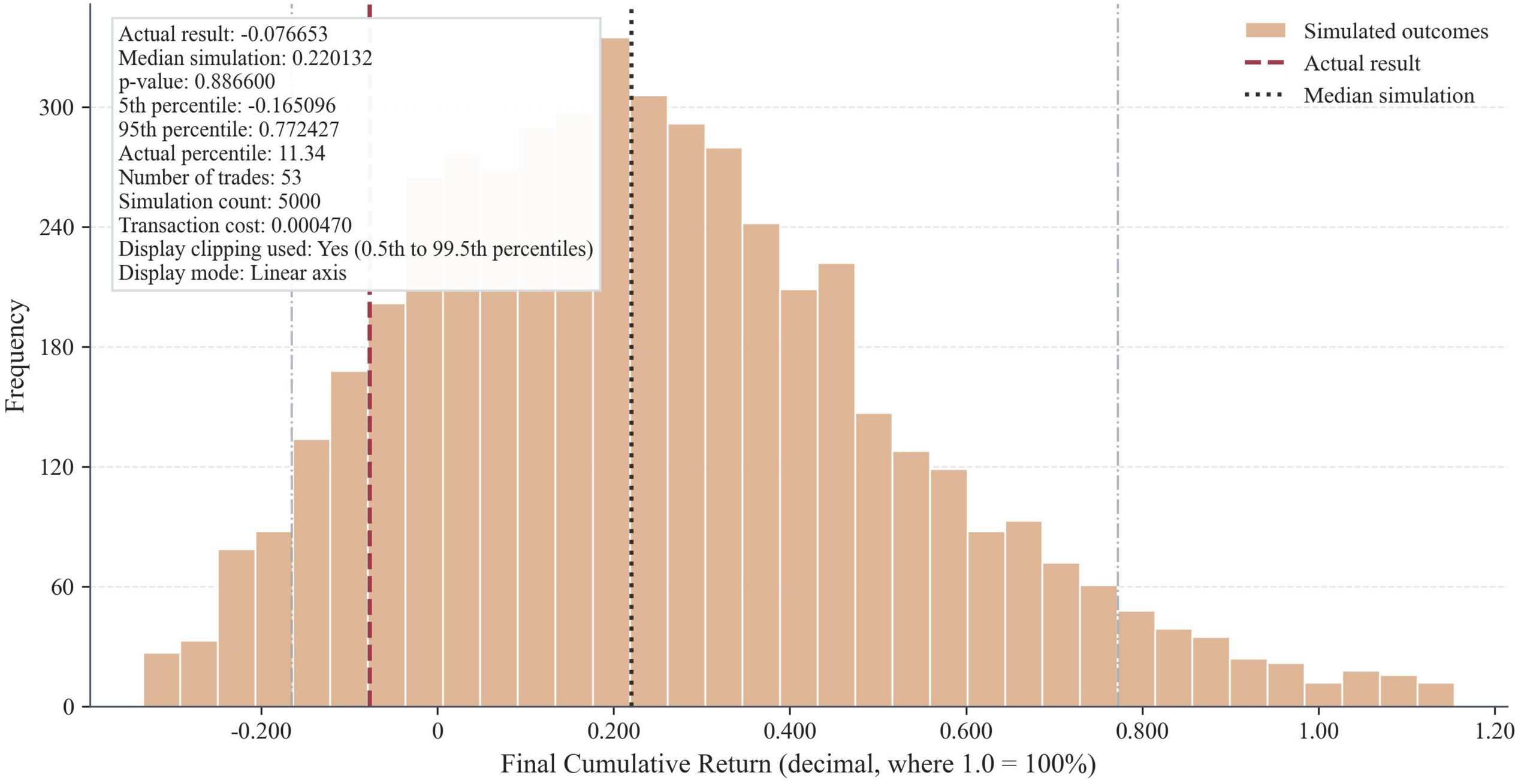
Trade-Level Return Ratio (unitless)

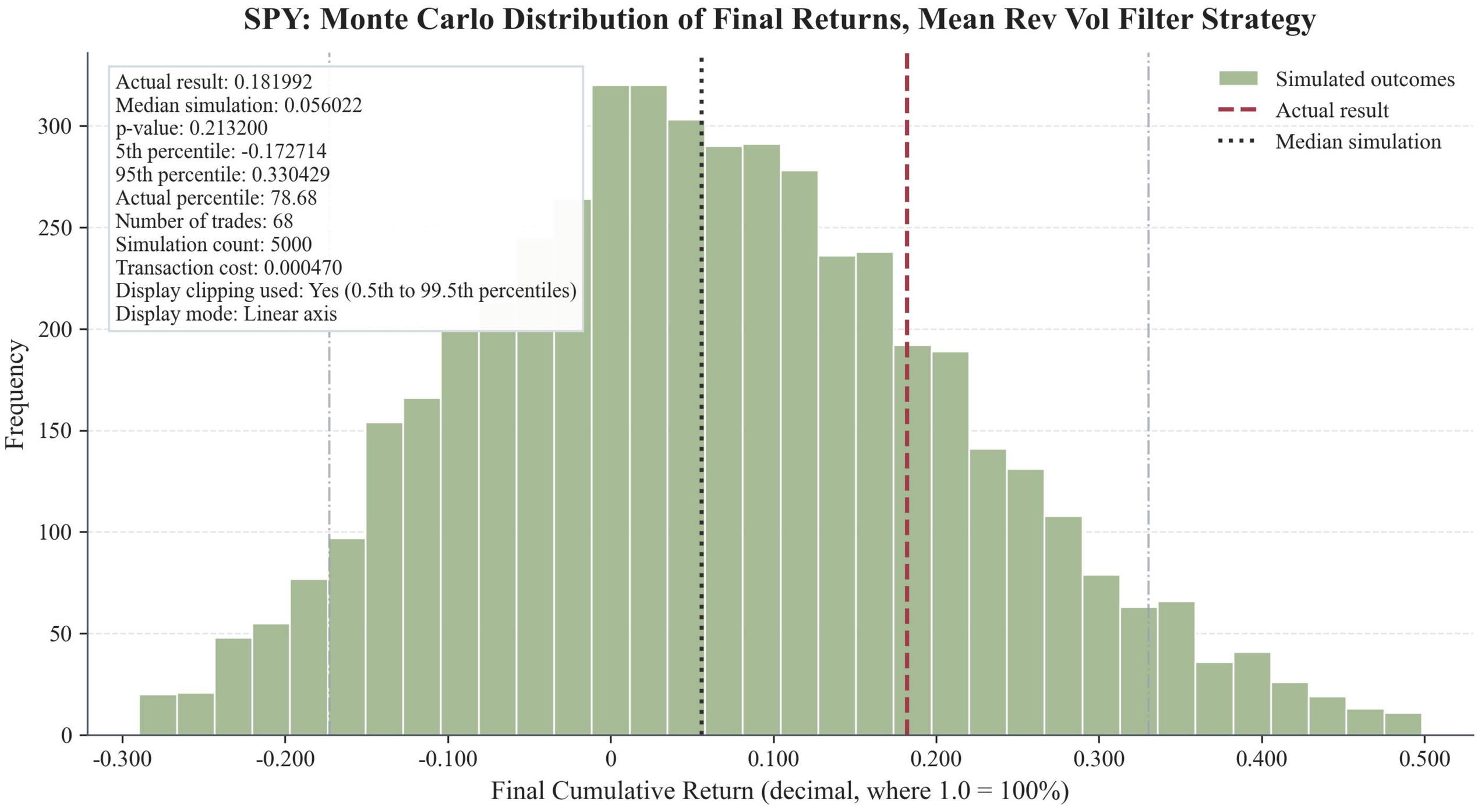
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3


SPY: Daily Equity Curves by Strategy and Benchmark
Trend Pullback
Breakout Vol+Mom
Mean Rev Vol Filter
Vol Managed TSMOM
Rel Strength Mom
Random
Buy & Hold
Cumulative Return (decimal, where 1.0 = 100%)
25.0
20.0
15.0
10.0
5.00
0
1996
2000
2004
2008
2012
2016
2020
2024
Date
Daily curves mark open positions to market using close prices. Buy and Hold is a passive benchmark.
Trend Pullback: 0.038215
Breakout Vol+Mom: -0.076653
Mean Rev Vol Filter: 0.181992
Vol Managed TSMOM: 1.695029
Rel Strength Mom: 0.188776
Random: -0.121924
Buy & Hold: 22.968554
Date span: 1994-11-10 to 2026-03-31

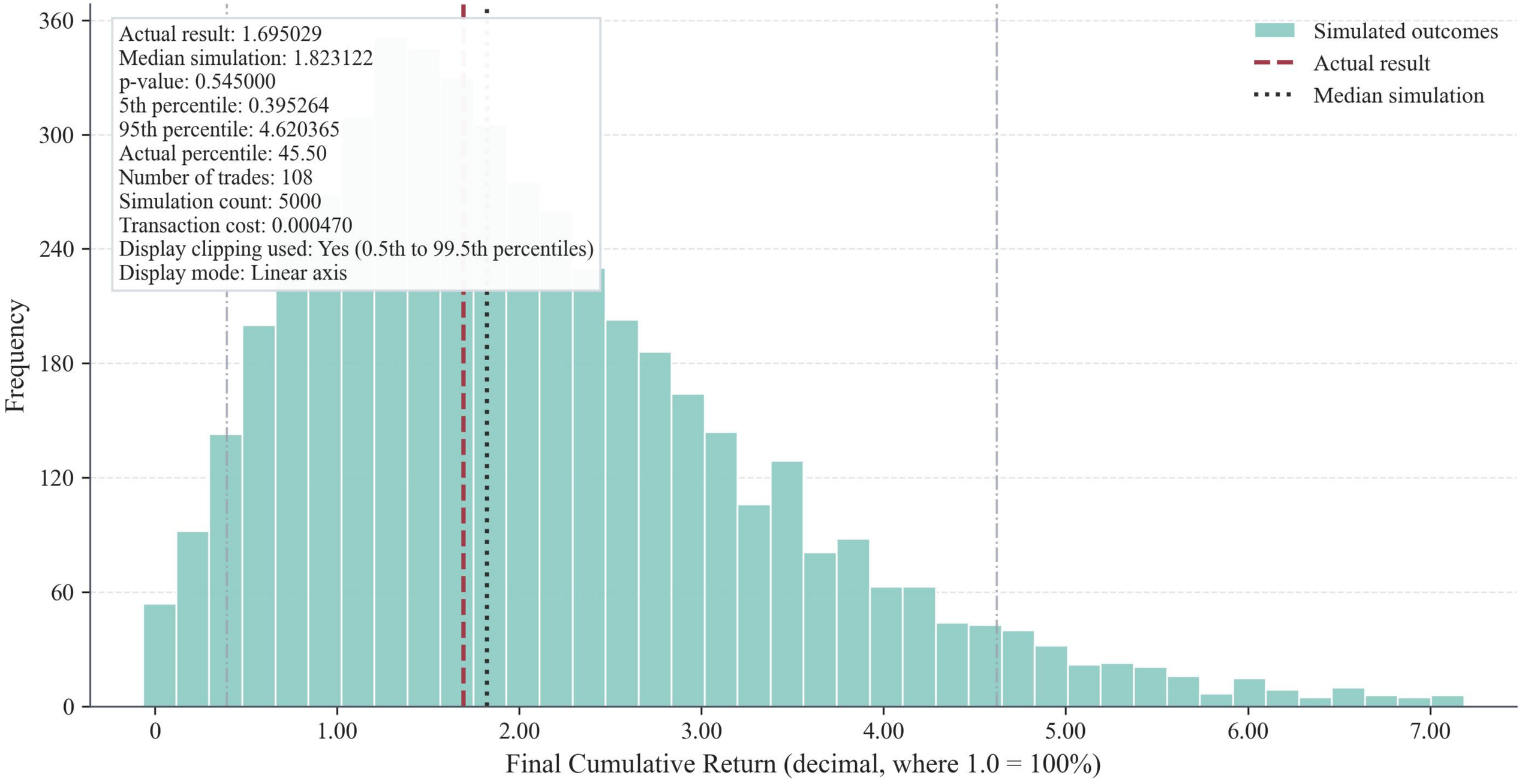

SPY: Monte Carlo Distribution of Final Returns, Trend Pullback Strategy
Actual result: 0.038215
Median simulation: 0.152995
p-value: 0.671600
5th percentile: -0.225528
95th percentile: 0.707651
Actual percentile: 32.84
Number of trades: 156
Simulation count: 5000
Transaction cost: 0.000470
Display clipping used: Yes (0.5th to 99.5th percentiles)
Display mode: Linear axis
Simulated outcomes
Actual result
Median simulation
Frequency
0
50
100
150
200
250
300
-0.400
-0.200
0
0.200
0.400
0.600
0.800
1.00
1.20
Final Cumulative Return (decimal, where 1.0 = 100%)

SPY: Monte Carlo Distribution of Final Returns, Breakout Vol+Mom Strategy

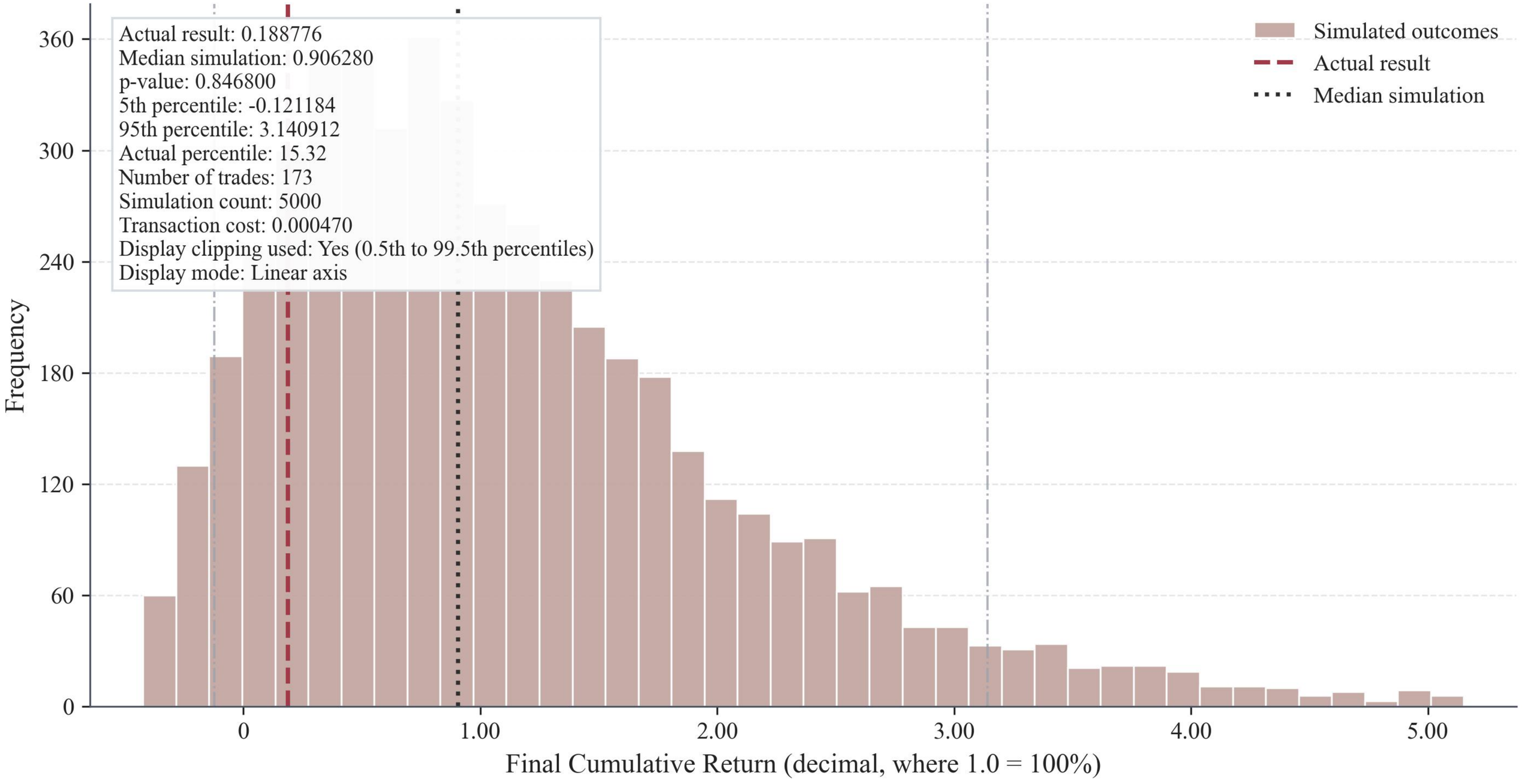



SPY: Monte Carlo Distribution of Final Returns, Mean Rev Vol Filter Strategy
Actual result: 0.181992
Median simulation: 0.056022
p-value: 0.213200
5th percentile: -0.172714
95th percentile: 0.330429
Actual percentile: 78.68
Number of trades: 68
Simulation count: 5000
Transaction cost: 0.000470
Display clipping used: Yes (0.5th to 99.5th percentiles)
Display mode: Linear axis
Simulated outcomes
Actual result
Median simulation
Frequency
0
50
100
150
200
250
300
-0.300
-0.200
-0.100
0
0.100
0.200
0.300
0.400
0.500
Final Cumulative Return (decimal, where 1.0 = 100%)

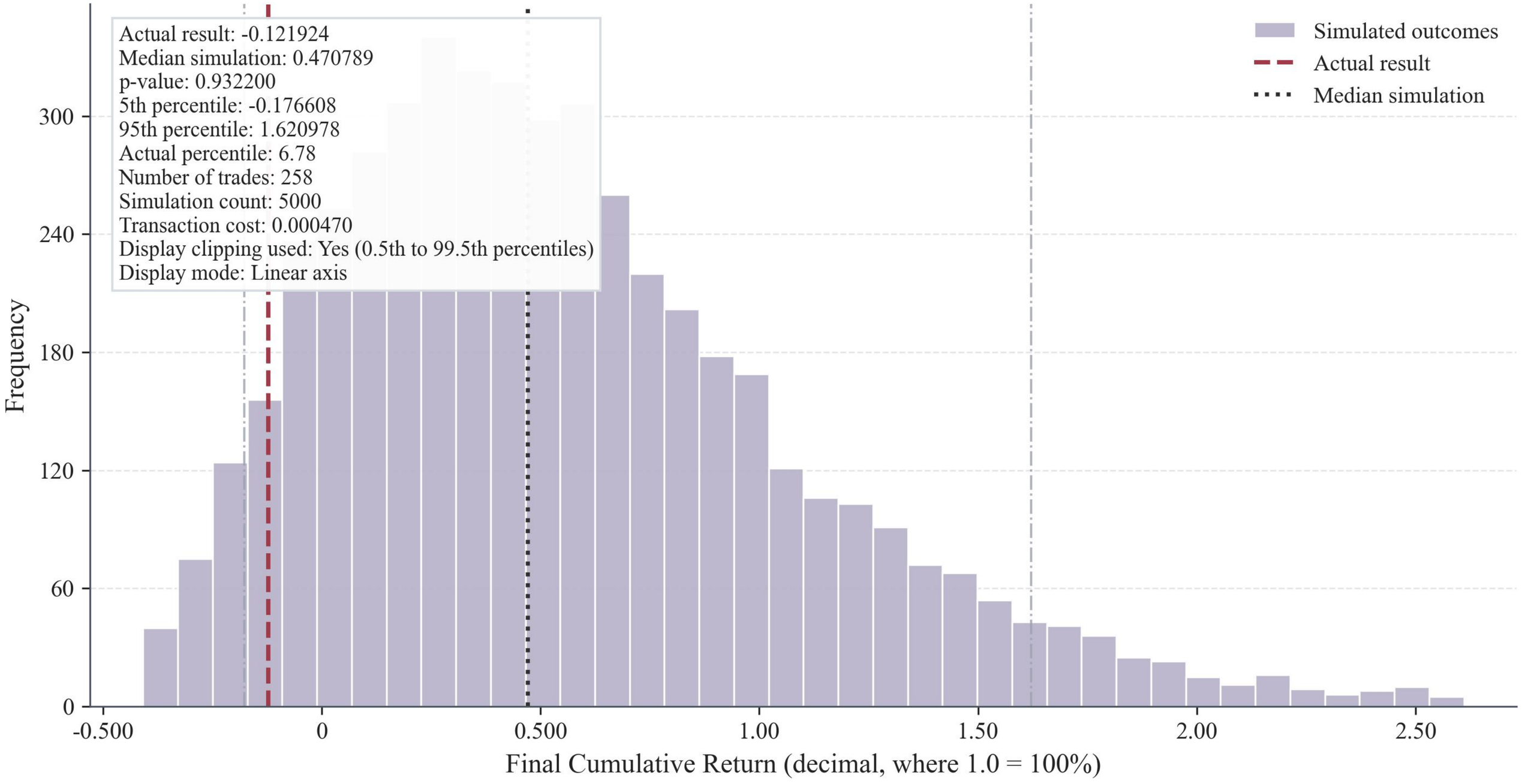
SPY: Monte Carlo Distribution of Final Returns, Vol Managed TSMOM Strategy

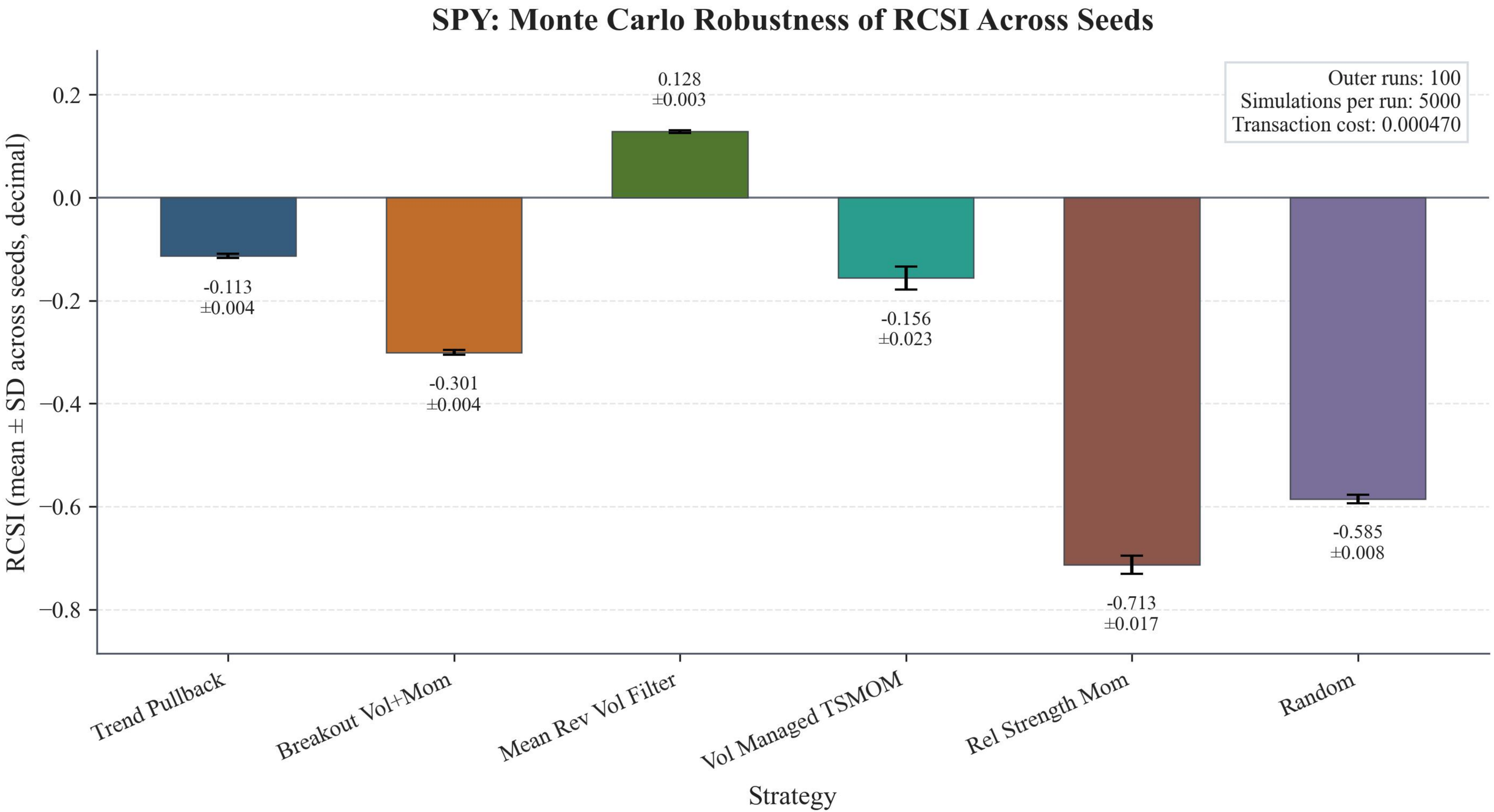


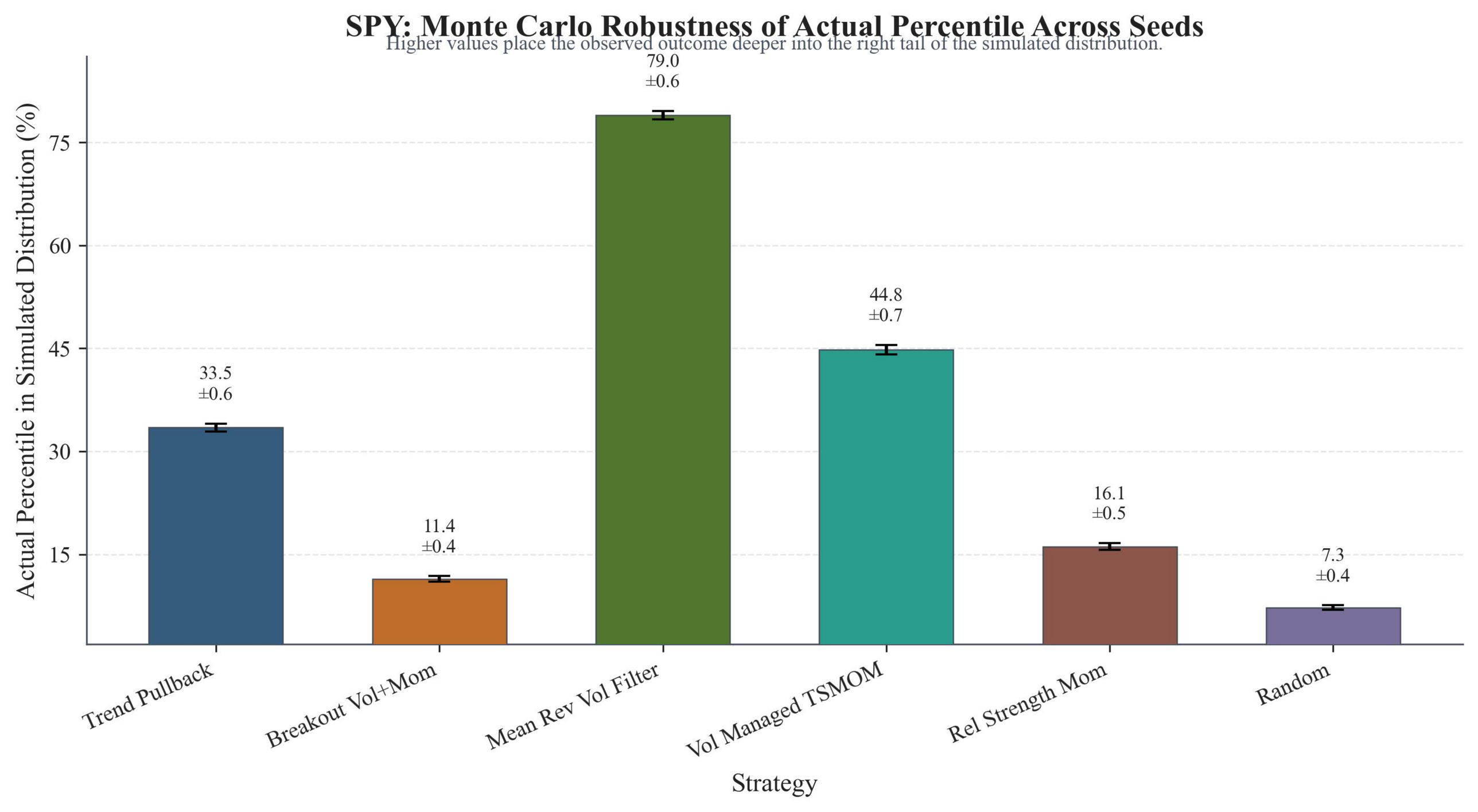
SPY: Monte Carlo Distribution of Final Returns, Rel Strength Mom Strategy

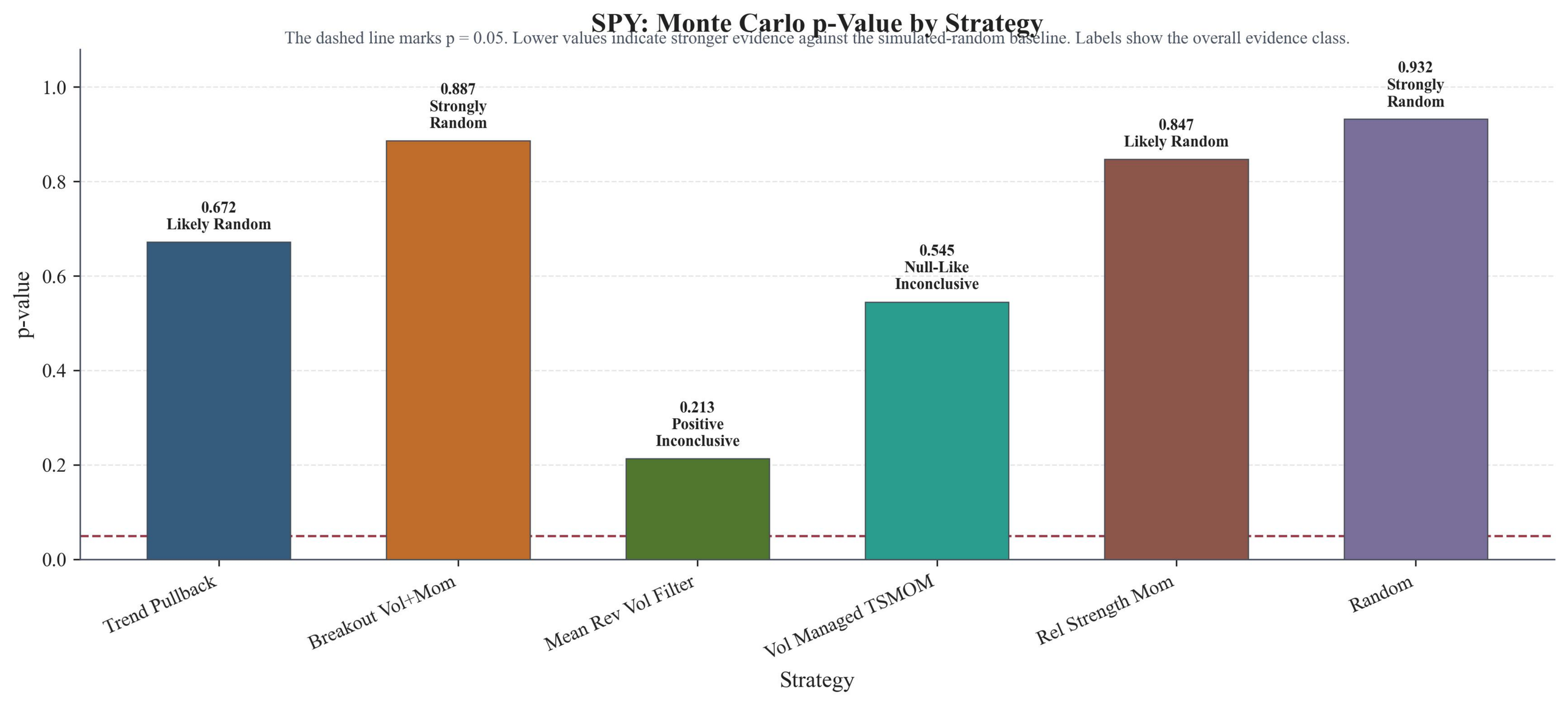


SPY: Monte Carlo Distribution of Final Returns, Random Strategy




SPY: Monte Carlo Robustness of RCSI Across Seeds
Outer runs: 100
Simulations per run: 5000
Transaction cost: 0.000470
RCSI (mean ± SD across seeds, decimal)
0.2
0.0
−0.2
−0.4
−0.6
−0.8
-0.113
±0.004
-0.301
±0.004
0.128
±0.003
-0.156
±0.023
-0.713
±0.017
-0.585
±0.008
Trend Pullback
Breakout Vol+Mom
Mean Rev Vol Filter
Vol Managed TSMOM
Rel Strength Mom
Random
Strategy


SPY: Monte Carlo Robustness of Actual Percentile Across Seeds
Higher values place the observed outcome deeper into the right tail of the simulated distribution.
Actual Percentile in Simulated Distribution (%)
75
60
45
30
15
33.5
±0.6
11.4
±0.4
79.0
±0.6
44.8
±0.7
16.1
±0.5
7.3
±0.4
Trend Pullback
Breakout Vol+Mom
Mean Rev Vol Filter
Vol Managed TSMOM
Rel Strength Mom
Random
Strategy


SPY: Monte Carlo p-Value by Strategy
The dashed line marks p = 0.05. Lower values indicate stronger evidence against the simulated-random baseline. Labels show the overall evidence class.
p-value
1.0
0.8
0.6
0.4
0.2
0.0
0.672
Likely Random
0.887
Strongly
Random
0.213
Positive
Inconclusive
0.545
Null-Like
Inconclusive
0.847
Likely Random
0.932
Strongly
Random
Trend Pullback
Breakout Vol+Mom
Mean Rev Vol Filter
Vol Managed TSMOM
Rel Strength Mom
Random
Strategy